Secure Elder-Care Monitoring Platform Video Based

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Project Proposal Report

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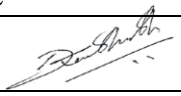
March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

- The rapid growth of the global elderly population has increased the demand for reliable home-care monitoring solutions. Elderly individuals who live alone or depend on caregivers often face risks such as falls, prolonged inactivity, or unsafe caregiver behavior. Traditional monitoring approaches mainly rely on wearable devices, sensor-based systems, or cloud-based video surveillance. However, these solutions present several limitations including privacy concerns, dependence on internet connectivity, limited behavioral analysis, and poor user compliance. Consequently, there is a need for intelligent and privacy-preserving monitoring systems capable of identifying abnormal situations in real time within home environments.
- This research focuses on developing a pose-based anomaly detection module designed to identify abnormal behaviors such as falls, aggressive actions, and prolonged inactivity. The main research gap lies in the lack of edge-based systems that analyze human motion while maintaining privacy and supporting real-time processing in elder-care environments. To address this gap, the proposed solution utilizes skeletal pose data extracted from video frames instead of storing or transmitting raw video footage.
- The system processes video input through pose estimation techniques to obtain joint coordinate data representing human body movements. These motion patterns are then analyzed using lightweight machine learning models, such as Long Short-Term Memory (LSTM) networks or autoencoder-based anomaly detection models, to learn normal behavior patterns and detect deviations. The entire detection pipeline is optimized for deployment on edge devices such as Webcam to ensure low latency and improved privacy protection.
- The expected outcome of this research is a real-time anomaly detection system capable of accurately identifying unsafe or unusual behaviors in elder-care settings. Socially, the system can improve elderly safety and provide reassurance to families. Commercially, it has potential applications in smart healthcare monitoring systems, home-care services, and intelligent surveillance solutions.

- **Keywords:** Anomaly Detection, Pose Estimation, Elder-Care Monitoring, Edge AI, Computer Vision

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
CV	Computer Vision
IoT	Photovoltaic
LSTM	Long Short-Term Memory
API	Application Programming Interface
SDG	Sustainable Development Goals
FPS	Frames Per Second
CPU	Central Processing Unit
GPU	Graphics Processing Unit
DSR	Design Science Research
YOLO	You Only Look Once
HCI	Human-Computer Interaction

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INTRODUCTION

1.1 Background and Context

- The global population is aging at an unprecedented rate, creating significant challenges for healthcare systems and families responsible for elderly care. Many elderly individuals prefer to live independently in their own homes rather than in assisted living facilities. However, aging often brings physical limitations, increased health risks, and greater dependence on caregivers. Situations such as falls, prolonged inactivity, and unsafe caregiver behavior can seriously threaten the safety and well-being of elderly individuals, especially when continuous supervision is not available. Therefore, reliable monitoring solutions are becoming increasingly important to ensure the safety of elderly people who receive home-based care.
- Several stakeholders are directly affected by this issue. Elderly individuals are the primary group at risk, as delayed detection of falls or emergencies can lead to severe injuries, hospitalization, or even fatal outcomes. Family members also experience anxiety and uncertainty when they cannot continuously monitor the well-being of their loved ones. Additionally, home-care service providers face challenges related to accountability and transparency, since there are limited mechanisms available to objectively verify caregiver activities and behavior in private home environments.
- From a global perspective, the number of people aged 60 years and above is rapidly increasing. According to international demographic reports, the elderly population is expected to double within the coming decades. In Sri Lanka, recent population statistics also indicate a steady rise in the aging population, which places additional pressure on healthcare infrastructure and caregiving services. This demographic shift highlights the urgent need for intelligent technological solutions that can assist in monitoring elderly individuals while maintaining dignity and privacy.
- Several technological approaches have been proposed to address this issue. Many existing elder-care monitoring systems rely on wearable devices, IoT sensors, or cloud-based surveillance systems. Although wearable devices can detect certain physiological signals or falls, they often suffer from poor user compliance because elderly individuals may forget or refuse to wear them. Sensor-based monitoring systems provide limited information about human behavior and are unable to accurately analyze complex activities. Cloud-based video surveillance solutions can perform advanced analytics, but they raise serious privacy concerns and depend heavily on stable internet connectivity and high computational resources.
- To overcome these limitations, this research proposes an **AI-driven, Video based elder-care monitoring platform** that uses computer vision and edge computing technologies. The system analyzes human body movements through skeletal pose estimation rather than storing raw video data. By extracting body joint coordinates from video frames, the system can identify behavioral patterns while preserving privacy. The platform integrates multiple intelligent modules including caregiver identity verification, movement tracking with geofencing, pose-based anomaly detection, and routine schedule monitoring. These

components work together to detect abnormal events such as falls, aggressive actions, or unusual inactivity in real time.

- The proposed solution also contributes to the **United Nations Sustainable Development Goals (SDGs)**, particularly **SDG 3: Good Health and Well-Being**, which focuses on improving health and safety for all age groups. Additionally, the project supports **SDG 11: Sustainable Cities and Communities** by promoting the use of intelligent technologies that enhance safety and quality of life for aging populations living independently.
- From a research perspective, this work belongs to the **Autonomous Intelligent Machines and Systems (AIMS)** research cluster, which focuses on the development of intelligent systems that combine artificial intelligence, computer vision, and embedded computing technologies to solve real-world problems. By integrating edge-based AI models with privacy-preserving monitoring techniques, this research aims to advance the development of smart healthcare monitoring solutions for elder-care environments.

1.2 Literature Review

Recent advancements in artificial intelligence, computer vision, and smart-healthcare systems have enabled the development of intelligent monitoring platforms for elderly care [1]. Many studies have explored wearable-based, sensor-based, and camera-based solutions to identify abnormal events such as falls, inactivity, or unusual behavior patterns. IoT-based elder-care systems are widely used. Kaluarachchi et al. proposed a smart elder-care framework that utilizes environmental sensors and wearable devices to monitor basic activity patterns and generate alerts during abnormal events [1]. However, IoT and wearable-based approaches often fail to analyze complex human behaviors. Computer vision offers powerful alternatives. Deep learning facial recognition models such as ArcFace provide highly discriminative facial embeddings for authentication tasks [2]. Pose-estimation frameworks like OpenPose enable real-time multi-person skeleton extraction, enabling privacy-preserving behavioral analysis [3]. Anomaly-detection research also contributes significantly. Hasan et al. introduced a deep temporal regularity learning model capable of detecting unusual activities in video sequences [4]. Motion-trajectory-based activity recognition methods, such as those proposed by Wang and Schmid, further improve recognition accuracy but require high computational resources [5]. Despite advancements, limitations remain: most systems focus on a single capability; many depend heavily on cloud-based processing; and conventional surveillance systems store raw video, raising privacy concerns. A key research gap is the lack of pose-based anomaly detection optimized for real-time edge deployment. This research addresses these gaps by developing a privacy-preserving, edge-based anomaly detection module that uses skeletal pose data and lightweight AI models. The system integrates facial verification, human tracking, and routine-monitoring components to create a comprehensive elder-care monitoring solution.

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1.3 Research Gap

- Although existing research has explored several approaches for monitoring elderly individuals using sensors, wearable devices, and computer vision technologies, significant gaps remain in the development of comprehensive and privacy-aware monitoring systems for home-care environments.

- Wearable- and IoT-based elder-care systems such as those proposed by Kaluarachchi et al. demonstrate that environmental sensing can detect basic activity patterns, yet they lack the ability to understand complex human motions and interactions [1]. These technologies also suffer from poor user compliance, as elderly individuals may forget to wear devices, remove them due to discomfort, or use them incorrectly, reducing system reliability.

- Current computer vision–based elder-care solutions also present limitations. Many focus on **single-task applications** such as fall detection, facial recognition, or movement tracking. For instance, deep face-recognition models like ArcFace excel at identity verification [2], and object-tracking frameworks like ByteTrack demonstrate strong multi-object tracking performance [6]. Similarly, OpenPose provides accurate real-time pose estimation [3]. However, most existing systems apply these models in isolation rather than integrating them into a unified elder-care monitoring framework. As a result, there is limited research addressing comprehensive behavioral monitoring—such as caregiver conduct, prolonged inactivity, or abnormal human–caregiver interactions.

- Another major limitation is the **heavy reliance on cloud-based video analytics**. Many video surveillance systems transmit raw video streams to remote servers for processing, which increases operational cost, introduces network latency, and raises significant privacy concerns associated with handling identifiable video footage. Studies on video-based anomaly detection, such as learning temporal regularity in video sequences [4] or unsupervised context-prediction methods [7], often focus on powerful but cloud-dependent models that are not optimized for resource-constrained edge environments.

- Furthermore, existing research rarely prioritizes **privacy-preserving behavioral analysis** using skeletal pose data instead of raw video frames. Although pose-estimation models like OpenPose [3] and trajectory-based activity-recognition approaches [5] provide valuable insights into human motion, they are not commonly applied within elder-care settings for privacy-focused monitoring. Pose-based representations remove identifiable visual details and are therefore more suitable for sensitive environments, yet their use in real-time anomaly detection for home-based elder care remains underexplored.

- Finally, recent surveys on human-action recognition and prediction [8] reveal that while deep learning has advanced activity understanding, only limited work focuses on **lightweight, edge-deployable anomaly detection** specifically designed for elderly-care safety. Most models require high computational capacity and are not optimized for embedded devices such as Raspberry Pi Or Webcam.

- To address these gaps, this research proposes a **privacy-preserving, edge-based anomaly detection module** that analyzes skeletal pose data extracted from video streams to detect abnormal behaviors such as falls, violent movements, and prolonged inactivity. By using pose-based motion features and integrating lightweight models guided by principles from LSTM-based temporal modeling [4] and unsupervised video-anomaly detection [7] the system aims to deliver accurate real-time monitoring on low-power edge devices. This approach reduces dependency on cloud infrastructure, strengthens privacy protection, and provides a scalable foundation for intelligent elder-care monitoring.

OBJECTIVE

2.1 Main Objective

The main objective of this research is to **design and develop a privacy-preserving, AI-driven elder-care monitoring system that operates on edge devices to verify caregiver identity, track human movement, and detect behavioral anomalies in real time, thereby improving the safety of elderly individuals and increasing transparency and accountability in home-care environments.**

2.2 Specific Objectives

- **To develop an edge-based facial verification module** capable of identifying authorized caregivers using deep learning face recognition models while ensuring that sensitive visual data is processed locally to preserve user privacy.
- **To implement a real-time human detection and tracking module** using computer vision techniques that can monitor the movement of individuals and detect entry or exit events within predefined zones around the home environment.
- **To design and train a pose-based anomaly detection model** that analyzes skeletal motion patterns to identify abnormal behaviors such as falls, aggressive actions, or prolonged inactivity in real time.
- **To develop a schedule monitoring and routine analysis module** that compares detected activities with predefined daily schedules in order to identify deviations such as missed meals or medication times.
- **To integrate all AI components into a unified edge-based monitoring system** deployed on a Webcam device to ensure low-latency processing, reduced cloud dependency, and improved privacy protection.
- **To evaluate the performance of the proposed system** using metrics such as detection accuracy, false-positive rate, and response time to validate its effectiveness in elder-care monitoring scenarios.

METHODOLOGY

3.1 Research Approach

- This research follows a **Design Science Research (DSR) approach**, which focuses on designing and evaluating a technological artifact that addresses a real-world problem. In this study, the artifact is an intelligent elder-care monitoring system that integrates multiple artificial intelligence models to detect abnormal situations in home environments. The design science approach is suitable because the objective of the research is not only to study a phenomenon but also to develop a practical solution that can improve safety and monitoring in elder-care scenarios.
- The research process involves several stages, including problem identification, system design, prototype development, model training, and experimental evaluation. First, the problem of limited supervision and safety risks in home-based elder-care environments is analyzed. Next, a conceptual system architecture is designed that integrates multiple computer vision modules. Afterward, individual components such as face verification, human tracking, anomaly detection, and schedule monitoring are developed and integrated into an edge-based platform. Finally, the system performance is evaluated through controlled experiments to assess accuracy, reliability, and real-time responsiveness.

3.2 Technical Methods and Algorithms

The proposed system uses a combination of **computer vision, machine learning, and edge computing techniques** to monitor activities and detect abnormal behaviors.

The first component is a **facial verification module** that identifies authorized caregivers. This module uses deep learning–based face recognition models such as Arc Face to extract facial embeddings from detected faces. The embeddings are compared with stored reference embeddings using similarity scoring to determine whether a person is authorized.

The second component is a **human detection and tracking module**. A lightweight object detection model such as YOLOv8 is used to detect people in video frames. The ByteTrack algorithm is applied to maintain consistent identity tracking of individuals across frames. This allows the system to monitor movement patterns and detect when a person enters or exits predefined zones within the home environment.

The third component, which represents the core focus of this research, is the **pose-based anomaly detection module**. MediaPipe Pose is used to extract skeletal joint coordinates from video frames, representing the position of body joints in two-dimensional space. These coordinates form a time-series representation of human motion. Machine learning models such as Long Short-Term Memory (LSTM) networks or autoencoder-based anomaly detection models are then trained to learn normal movement patterns. When a significant deviation from learned patterns is detected, the system classifies the event as an anomaly, such as a fall, aggressive movement, or prolonged inactivity.

The final component is a **schedule monitoring module** that compares detected behaviors with predefined daily routines. Rule-based logic is used to identify deviations such as missed meals or medication times. Alerts are generated when irregularities are detected.

3.3 Tools and Platforms with Justification

The system is implemented using several software tools and hardware platforms that support efficient development and deployment of machine learning models.

Python is used as the primary programming language because it provides extensive support for machine learning and computer vision libraries. **OpenCV** is used for video processing and image manipulation, while **MediaPipe** is utilized for efficient real-time pose estimation.

Deep learning models are developed using frameworks such as **TensorFlow or PyTorch**, which provide robust support for training and deploying neural networks. The **Ultralytics YOLOv8 framework** is used for person detection due to its balance between detection accuracy and computational efficiency.

The system is deployed on a **Raspberry Pi edge device or Logitech C270 , Logitech C310**

which allows local processing of video data. Edge computing reduces the need for cloud-based processing, minimizes network latency, and improves privacy by keeping sensitive data within the local environment.

For system integration and communication between components, lightweight backend frameworks such as **FastAPI** may be used. Additionally, **MongoDB or lightweight local storage** is used to store embeddings, logs, and event records securely.

3.4 Data Collection and Ethical Considerations

The data required for training and evaluating the anomaly detection component will primarily be obtained from **publicly available human activity recognition and fall detection datasets**. These datasets include labeled video or skeletal motion data representing normal human activities such as walking, sitting, and bending, as well as abnormal behaviors such as falls or sudden aggressive movements. Platforms such as **Kaggle** provide several curated datasets that can be used to train and benchmark anomaly detection models in surveillance and activity recognition research.

Instead of developing a model entirely from scratch, this research will utilize **pre-trained machine learning models** and fine-tune or optimize them for the specific context of elder-care monitoring. Transfer learning techniques will be applied to adapt the pre-trained models so they can accurately recognize abnormal behaviors relevant to home-care environments. This approach reduces the amount of training data required while improving computational efficiency for deployment on edge devices.

To maintain privacy and ethical standards, the system will process motion information using **pose estimation techniques that extract skeletal joint coordinates** from video frames. Only the derived skeletal data will be used for training and evaluation, ensuring that

identifiable visual information is not stored or processed unnecessarily. Since the datasets used are publicly available and already anonymized for research purposes, no personally identifiable information will be collected or stored during the study.

All datasets will be used in accordance with their respective licenses and research guidelines. This ensures responsible data usage while allowing the proposed system to be trained and evaluated using reliable and widely recognized benchmark datasets.

3.5 Validation Strategy and Performance Metrics

The performance of the proposed system will be evaluated through experimental testing in simulated home-care environments. Several performance metrics will be used to assess the effectiveness of the anomaly detection and monitoring modules.

For the anomaly detection model, evaluation metrics such as **accuracy, precision, recall, and F1-score** will be used to measure classification performance. The **false positive rate** will also be analyzed to ensure that the system does not generate excessive incorrect alerts.

Additionally, **detection latency** will be measured to evaluate how quickly the system can identify abnormal events in real time. Since the system operates on edge hardware, computational efficiency and processing speed will also be analyzed.

The system will be tested under different environmental conditions, including variations in lighting, camera angles, and movement speeds, to evaluate robustness. Through these experiments, the research aims to demonstrate that the proposed system can effectively detect abnormal situations while maintaining privacy and real-time performance.

3.6 Evaluation Plan

The evaluation plan outlines the systematic procedures that will be used to assess the effectiveness, accuracy, and real-time performance of the proposed elder-care monitoring system. This plan ensures that each functional component face verification, human tracking, pose-based anomaly detection, and schedule monitoring is tested under controlled and realistic conditions to determine whether it meets the system requirements.

3.6.1 Evaluation Environment

Testing will be conducted in a simulated home-care environment that resembles typical residential conditions, including variations in:

- Indoor lighting (bright, dim, natural light)
- Camera placement (high angle, eye-level angle)
- Object occlusions and movement variability
  This ensures that evaluation results reflect real-world usage scenarios.

3.6.2 Component Evaluation Procedures

Pose-Based Anomaly Detection Module

- Test using both public datasets and real-world recordings in the simulated environment.
- Measure:
    - Anomaly classification accuracy
    - F1-score for fall/aggressive movement detection
    - False positives and false negatives
    - Latency (time from event to detection)
      This component will undergo stress testing with fast movements, occlusions, and varied motion speeds.

3.6.3 Success Criteria

The system will be considered successful if it meets the following thresholds:

- ≥ 85% anomaly detection accuracy with low false alerts
- ≤ 1.5 seconds detection-to-alert latency
- Positive usability feedback from test participants

LIMITATIONS AND CHALLENGES

Despite the potential of the proposed elder-care monitoring system, several limitations and challenges may impact system performance, deployment feasibility, and user acceptance. These factors are important to acknowledge as they help define the boundaries of the research and identify areas for future improvement.

1. Hardware and Computational Constraints

Since the system is designed to operate on edge devices such as Raspberry Pi or resource-limited webcams, the computational power available for real-time processing is restricted.
Lightweight models may need to compromise on accuracy, particularly for complex activities or fast movements.

2. Sensitivity to Environmental Conditions

Computer vision systems are highly dependent on the surrounding environment.
Variations in lighting, background clutter, camera angles, and occlusions can reduce detection and pose-estimation accuracy.
Real-world home environments may not always provide ideal conditions for consistent performance.

3. Data Limitations for Anomaly Detection

Anomaly detection models require diverse and well-annotated data representing falls, aggressive actions, and prolonged inactivity.
Public datasets may not perfectly match the home-care scenarios targeted in this research.
Collecting custom data with elderly participants raises ethical and privacy constraints.

4. False Positives and False Negatives

Although the system aims to detect abnormal events, errors are possible.
False positives can cause unnecessary panic or notification overload, while false negatives could delay responses to real emergencies.
Balancing sensitivity and reliability remains a challenge.

5. Privacy and Ethical Constraints

Even though the solution uses pose-based representations rather than raw video, some users may remain uncomfortable with camera-based monitoring.
Ensuring data protection, secure storage, and transparent usage policies is essential to build user trust.

6. Deployment and User Acceptance Challenges

Adoption requires acceptance from elderly individuals, family members, and caregivers.
Some stakeholders may be hesitant to rely on automated systems or may lack technical literacy.
Proper onboarding and user-friendly interfaces are required to ensure widespread usability.

7. Real-Time Performance and Latency

Maintaining low detection-to-alert latency across all modules simultaneously is challenging, especially on low-power devices.
High-resolution video or multi-person tracking may further increase processing time.

HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of Proposed Solution and Components

The proposed system is an **AI-driven elder-care monitoring platform** designed to enhance elderly safety and caregiver accountability through real-time video analysis. The system operates on an **edge computing architecture using a Web cam device**, ensuring that sensitive video data is processed locally to protect user privacy and reduce network dependency.

The architecture consists of **four main intelligent components** that operate collaboratively:

1. **Face Verification Module** – verifies the identity of caregivers entering the monitored environment using facial recognition techniques.
2. **Human Detection and Tracking Module** – detects and tracks individuals within the monitored area using computer vision models such as YOLO and ByteTrack.
3. **Pose-Based Anomaly Detection Module** – analyzes skeletal movement patterns to detect abnormal behaviors such as falls, aggression, or prolonged inactivity.
4. **Schedule Monitoring Module** – compares detected behaviors with predefined daily routines to identify deviations such as missed medication or meals.

All modules process data locally on the edge device and generate alerts when abnormal conditions are detected. Notifications are then delivered to family members or caregivers through a monitoring dashboard or mobile alert system.

4.2 System Diagram of Components

Below is the **high-level architecture flow**.

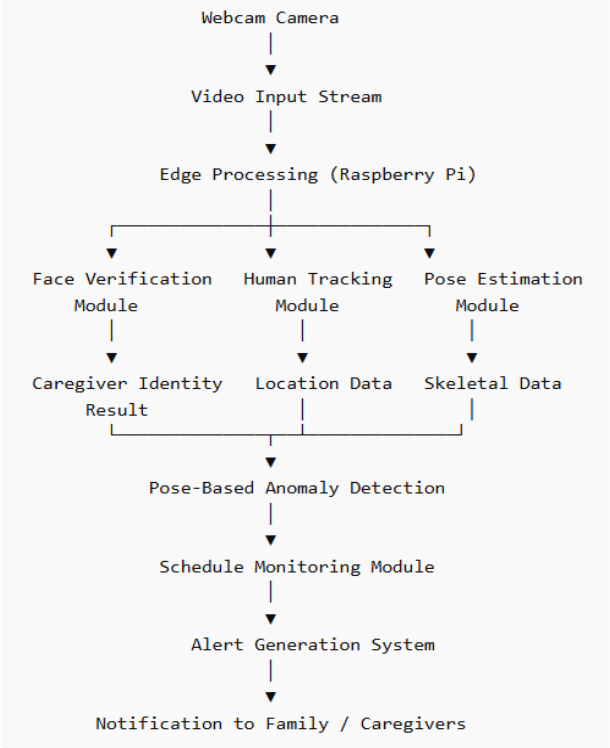


Figure 1: High level Architecture

4.3 Individual Component

The **pose-based anomaly detection module** is responsible for identifying abnormal human behavior using skeletal motion data.

Process steps include:

1. Capturing live video frames from the webcam.
2. Applying **Media Pipe Pose** to extract skeletal joint coordinates.
3. Converting skeletal data into motion sequences.
4. Feeding the motion sequences into a trained anomaly detection model.
5. Classifying behaviors such as:
    - Falls
    - Aggressive movements
    - Prolonged inactivity
6. Triggering alerts when abnormal patterns are detected.

This approach ensures **privacy preservation**, as the system processes only skeletal motion data rather than storing raw video frames.

4.4 Component Interactions

Each module in the system communicates through structured data outputs.

- The **face verification module** identifies authorized caregivers.
- The **tracking module** monitors movement and zone activity.
- The **anomaly detection module** analyzes human behavior patterns.
- The **schedule monitoring module** compares detected actions with predefined routines.

The outputs of these modules are integrated to provide a comprehensive monitoring solution that detects both behavioral anomalies and routine deviations.

4.5 Data Flow Explanation

The system follows a structured **input–process–output pipeline**:

Input:

- Live video feed from webcam.

Processing:

- Face verification
- Person detection and tracking
- Pose estimation
- Anomaly detection
- Schedule analysis

Output:

- Real-time alerts
- Event logs
- Notifications to family members or administrators.

4.6 Scalability and Security Considerations

To ensure scalability and security, the system is designed with the following principles:

- **Edge-based processing** reduces dependency on cloud infrastructure and improves response time.
- **Privacy-preserving pose estimation** prevents storage of identifiable video data.
- **Lightweight AI models** ensure compatibility with low-power edge devices such as Raspberry Pi.
- **Secure event logging and alert mechanisms** protect sensitive monitoring data.

This architecture enables the system to scale to multiple households while maintaining strong privacy and security protections.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

Several stakeholders interact with or benefit from the proposed elder-care monitoring system. Identifying these stakeholders helps define the system requirements and ensures the solution addresses real-world needs.

Elderly Individuals – Elderly people are the primary users of the system. The system aims to improve their safety by detecting dangerous situations such as falls, prolonged inactivity, or unsafe interactions.

Family Members / Guardians – Family members require reliable monitoring of their loved ones when they are not physically present. The system provides alerts and activity reports to help them stay informed about the well-being of the elderly individual.

Caregivers – Authorized caregivers interact with elderly individuals and perform daily support activities. The system verifies caregiver identity and monitors movements to ensure safe and responsible caregiving.

Healthcare Service Providers / Care Agencies – Care organizations can use the system to ensure accountability and maintain service quality by monitoring caregiver activities and responding to incidents when necessary.

System Administrators – Administrators are responsible for system configuration, managing authorized users, maintaining system updates, and ensuring the system operates correctly.

5.2 Functional Requirements

The system must provide several functional capabilities to achieve its objectives.

1. The system shall capture live video streams from a camera installed within the home environment.

2. The system shall verify the identity of caregivers using facial recognition techniques to ensure that only authorized individuals interact with the elderly person.
3. The system shall detect and track human movement within the monitored environment using computer vision algorithms.
4. The system should define and monitor restricted or predefined zones to detect entry or exit events within the home environment.
5. The system analyze human body movements using pose estimation techniques to detect abnormal behaviors such as falls, aggressive movements, or prolonged inactivity.
6. The system should generate real-time alerts when abnormal or unsafe situations are detected.
7. The system should monitor predefined daily schedules and detect deviations such as missed meals or medication times.
8. The system shall store event logs and system alerts for later review by authorized users.
9. The system should provide notifications to family members or administrators when critical incidents occur.

5.3 Non-Functional Requirements

Non-functional requirements define the quality attributes of the system, including performance, security, and usability.

- **Security**
  The system must ensure that sensitive data is protected from unauthorized access. Facial embeddings, activity logs, and system data should be securely stored, and access to system controls must be restricted to authorized users.
- **Privacy**
  The system must minimize the storage of raw video data and instead process anonymized skeletal pose information whenever possible to preserve user privacy.
- **Performance**
  The system should process video frames and detect abnormal behaviors in near real time. The detection latency should remain low to ensure timely alerts in emergency situations.
- **Reliability**
  The system must operate continuously and provide consistent monitoring without frequent failures or interruptions.

- **Scalability**
  The system should be designed in a modular manner so that additional monitoring components or sensors can be integrated in the future.
- **Usability**
  The system interface and alerts should be simple and easy to understand so that family members and caregivers can quickly interpret notifications and system status.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

- The primary target market for the proposed elder-care monitoring system includes **families with elderly individuals who require continuous supervision**, particularly those who receive home-based caregiving services. As the aging population continues to grow globally, many families prefer home-care solutions rather than institutional care facilities. This creates an increasing demand for affordable monitoring technologies that can ensure the safety and well-being of elderly individuals.
- In addition to families, **home-care agencies and healthcare service providers** represent another significant market segment. These organizations can use the system to improve caregiver accountability, enhance patient safety, and provide transparent monitoring services to clients.
- The typical customer persona for this product would be **middle-income households with elderly parents living alone or receiving caregiver assistance**, as well as **private healthcare providers and home nursing services** seeking reliable monitoring tools. The system can also appeal to **assisted living facilities and retirement communities** that require automated monitoring to enhance resident safety.

6.2 Value Proposition

- The proposed solution offers a **privacy-preserving, AI-driven monitoring platform** designed specifically for elder-care environments. Unlike traditional surveillance systems that continuously record video footage, the proposed system analyzes **skeletal pose data extracted from video frames**, reducing privacy concerns while still allowing accurate detection of abnormal behaviors.
- The system provides several key benefits:
  - Real-time detection of falls, abnormal movements, and inactivity
  - Verification of caregiver identity to prevent unauthorized access
  - Monitoring of caregiver movements and activity zones
  - Detection of deviations from daily schedules such as missed meals or medication
  - Edge-based processing that reduces cloud dependency and improves response time
- These features provide families with **peace of mind**, improve **elder safety**, and enable **care agencies to maintain higher service transparency and accountability**.

6.3 Revenue Model

- The proposed system can adopt a **hybrid revenue model** that combines hardware sales with subscription-based services.
- One potential approach is to sell the **hardware package**, which includes the Raspberry Pi device, camera module, and pre-installed monitoring software. Customers would make a one-time payment to purchase and install the system in their homes.

- In addition, the system could offer **subscription-based services** that include advanced monitoring features such as mobile notifications, remote monitoring dashboards, data analytics, and system updates. Monthly or annual subscription plans can provide continuous technical support and software improvements.
- This combination of hardware sales and recurring service subscriptions can create a **sustainable long-term revenue stream** for the product.

6.4 Cost Estimation

- The estimated cost of implementing the system includes both hardware and development components.
- Key hardware components include:
  - Raspberry Pi or Webcam edge computing device
  - Camera module for video capture
  - Storage devices and power supply
- These components are relatively low-cost and widely available, making the system affordable for home users. Additional costs may include system development, software maintenance, and cloud infrastructure for optional remote services.
- By using **edge computing and open-source machine learning frameworks**, the overall operational cost can be significantly reduced compared to traditional cloud-based monitoring systems.

6.5 Pricing Strategy

- The pricing strategy aims to make the system **affordable for middle-income households while maintaining profitability**.
- A potential pricing model may include:
  - **Initial hardware package**: approximately $120–$200 depending on hardware configuration
  - **Monthly subscription plan**: $5–$15 per month for monitoring services, software updates, and technical support
- For healthcare organizations or elder-care agencies, a **bulk pricing model** or enterprise package could be offered to support multiple monitoring units across several locations.
- This tiered pricing strategy allows the product to reach both **individual households and institutional clients**.

6.6 Competitive Advantage

- The proposed system offers several competitive advantages compared to existing elder-care monitoring solutions.
- First, the system is designed with a **privacy-preserving architecture** that uses skeletal pose data rather than storing raw video footage. This approach reduces privacy concerns and makes the system more suitable for home environments.
- Second, the system uses **edge computing**, allowing all critical processing to occur locally on the device. This reduces network dependency, improves response time, and lowers operational costs compared to cloud-based solutions.
- Third, the platform integrates **multiple monitoring capabilities within a single system**, including caregiver verification, human tracking, anomaly detection, and schedule monitoring. Many existing solutions focus only on a single feature such as fall detection, whereas this system provides a more comprehensive monitoring framework.
- Finally, the use of **lightweight AI models optimized for edge devices** makes the system scalable and energy efficient, allowing it to be deployed in a wide range of residential and healthcare environments.

7. BUDGET AND JUSTIFICATION

7.1 Hardware Costs

Table 1:Hardware Costs

Item	**Description**	**Estimated Cost (USD)**
Raspberry Pi 4 (4GB)	Edge computing device used to run AI models and process video streams locally	$70
Logitech C270 Webcam	Popular low-cost webcam commonly used in computer vision and AI experiments	$12
Logitech C920 Webcam (Optional)	Higher quality webcam with improved image clarity for better pose detection	$10
Generic 1080p USB Webcam (Alternative)	Budget-friendly camera option suitable for basic testing	$15
MicroSD Card (64GB)	Storage for operating system, datasets, and model files	$12
Power Supply for Raspberry Pi	Required to power the edge device	$10
Miscellaneous Accessories	Cables, mounting support for camera setup	$10

Estimated Total Hardware Cost: $100 – $150

(The cost varies depending on the camera model used.)

7.2 Software Licenses

The project mainly utilizes **open-source software tools and frameworks**, which significantly reduces development costs.

Software	**Purpose**	**Cost**
Python	Main programming language	Free
OpenCV	Image processing and video analysis	Free
MediaPipe	Human pose estimation framework	Free

Software	Purpose	Cost
TensorFlow / PyTorch	Deep learning model training and inference	Free
Ultralytics YOLOv8	Human detection and tracking	Free (open-source version)
FastAPI	Backend API development	Free

Table 2: Software License cost

Estimated Software Cost: $0

Using open-source technologies makes the system **economically viable for research and future scalability**.

7.3 Cloud Services

The system is designed to run primarily on **edge computing devices**, meaning that most processing is performed locally on the Raspberry Pi Or Webcam. This significantly reduces the need for cloud infrastructure.

However, minimal cloud resources may be used for:

- Dataset downloading and storage
- Development testing
- Optional remote monitoring features

Service	Purpose	Estimated Cost
Kaggle / Public datasets	Training datasets	Free
Google Colab	Model experimentation and training	Free tier
Optional Cloud Storage	Backup or experiment storage	$5 (estimated)

Table 3: Cloud services cost

Estimated Cloud Cost: $0 – $5

7.4 Human Effort Cost

Since this is an **academic research project conducted by undergraduate students**, no direct financial cost is associated with development work. The system will be designed, implemented, and tested by the research team as part of the final-year project requirements.

However, the project involves significant **time and technical effort**, including:

- AI model development and optimization
- System integration and testing
- Dataset preprocessing and evaluation
- Documentation and report preparation

7.5 Justification for Expenses

The selected hardware and software components are chosen to **minimize cost while maintaining sufficient computational capability for AI-based video analysis**.

The **Raspberry Pi 4** is selected because it supports edge-based machine learning applications and has sufficient processing power to run lightweight AI models. The **Logitech C270 webcam** is proposed as the primary camera due to its affordability and widespread use in AI and computer vision projects. The **Logitech C920** or a **generic 1080p webcam** can be used as alternative options when higher video quality is required for improved pose estimation accuracy.

Using **open-source machine learning frameworks** such as TensorFlow, PyTorch, MediaPipe, and OpenCV eliminates licensing costs while providing powerful tools for developing AI-based systems.

The reliance on **edge computing instead of cloud-based processing** further reduces operational costs while improving system privacy and response speed.

Overall, the proposed budget demonstrates that the system can be implemented using **affordable hardware and freely available software tools**, making it suitable for both academic research and potential real-world deployment.

Algorithms / Models to Be Used – Pose-Based Anomaly Detection

The proposed anomaly detection component will use **MediaPipe Pose** to extract skeletal joint coordinates from each video frame and convert them into **time-series motion sequences**. These sequences will be preprocessed through normalization, smoothing, and temporal windowing to improve robustness against camera angle, scale, and movement speed variations. A **lightweight LSTM network** will then be trained to classify motion patterns into normal activity, falls, aggressive actions, and prolonged inactivity, since LSTM is effective at learning temporal dependencies in sequential movement data. In parallel, an **Autoencoder-based anomaly detection model** will be trained on normal activity patterns and will use **reconstruction error** to detect unusual movement sequences. A hybrid decision mechanism combining the LSTM output and anomaly score will improve detection reliability and reduce false positives. For prolonged inactivity, an additional rule-based temporal threshold will be applied to detect extended low-motion periods. Finally, the trained models will be optimized for **real-time deployment on Raspberry Pi edge devices**, enabling low-latency, privacy-preserving anomaly detection without storing or transmitting raw video footage.

WORK BREAKDOWN STRUCTURE (WBS)

The Work Breakdown Structure (WBS) defines the major tasks and responsibilities assigned to each project member. The tasks are distributed to ensure balanced workload among team members while aligning with the overall project timeline. Each component contributes to the development of the AI-driven elder-care monitoring system.

8.1 Task Breakdown per Member

Member 1 – Face Verification Module

Responsible for developing the caregiver identity verification system.

Tasks:

- Configure Raspberry Pi environment for computer vision processing
- Integrate webcam (Logitech C270 / C920 / generic 1080p webcam) for video capture
- Implement face detection and facial embedding extraction
- Integrate pre-trained ArcFace model for caregiver verification
- Store authorized caregiver embeddings securely
- Develop similarity comparison logic for identity verification
- Generate alerts for unauthorized access

Member 2 – Human Detection and Tracking Module

Responsible for monitoring movement and zone violations.

Tasks:

- Install and configure YOLOv8 object detection model
- Implement person detection using camera input
- Integrate ByteTrack tracking algorithm
- Develop zone configuration system for monitoring areas
- Detect entry and exit events within defined zones
- Generate alerts for restricted zone violations

Member 3 – Pose-Based Anomaly Detection Module

Tasks:

- Select suitable pre-trained human activity recognition models
- Obtain activity recognition datasets from Kaggle
- Preprocess skeletal motion datasets for training
- Apply MediaPipe Pose to extract skeletal coordinates from video frames
- Fine-tune anomaly detection model for fall detection and abnormal activity recognition
- Optimize model for edge deployment on Pi or webcam
- Evaluate model accuracy and detection performance

Member 4 – Schedule Monitoring Module

Tasks:

- Design routine schedule database for elderly daily activities
- Define rule-based logic for activity validation
- Receive activity data from detection modules
- Compare detected behavior with scheduled routines
- Identify deviations such as missed meals or medication
- Generate alerts and daily reports

8.2 Timeline Alignment

The project tasks are organized according to the research timeline to ensure systematic progress.

Phase	**Activities**
Phase 1	Literature review and requirement analysis
Phase 2	System architecture design
Phase 3	Development of individual modules
Phase 4	Model training and optimization

Phase	Activities
Phase 5	Integration of all system components
Phase 6	Testing and performance evaluation
Phase 7	Documentation and final report preparation

Table 4:Timeline Alignment Phases

8.3 Balanced Workload Distribution

The project workload is distributed evenly among team members to ensure efficient collaboration. Each member is responsible for a specific system component while also contributing to integration, testing, and documentation activities.

This modular development approach allows team members to work independently on their respective components while maintaining coordination during system integration. Shared responsibilities such as dataset preparation, system testing, and report writing ensure equal participation across the research team.

WBS Diagram (Visual Representation)

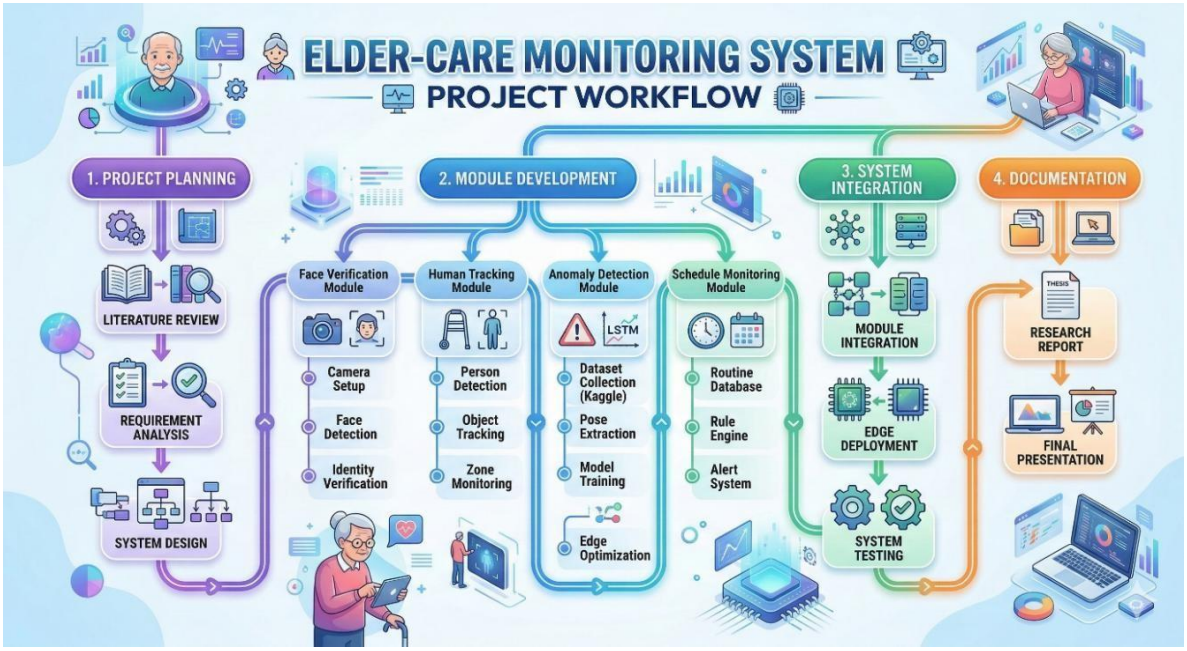


Figure 2: WBS DIAGRAM

9. GANTT CHART

The following Gantt chart presents the timeline for the development of the AI-based elder-care monitoring system. The project tasks are organized according to major research phases, including literature review, system design, model development, system integration, and evaluation. Key academic milestones such as proposal presentation, progress evaluations, and final assessment are also included.

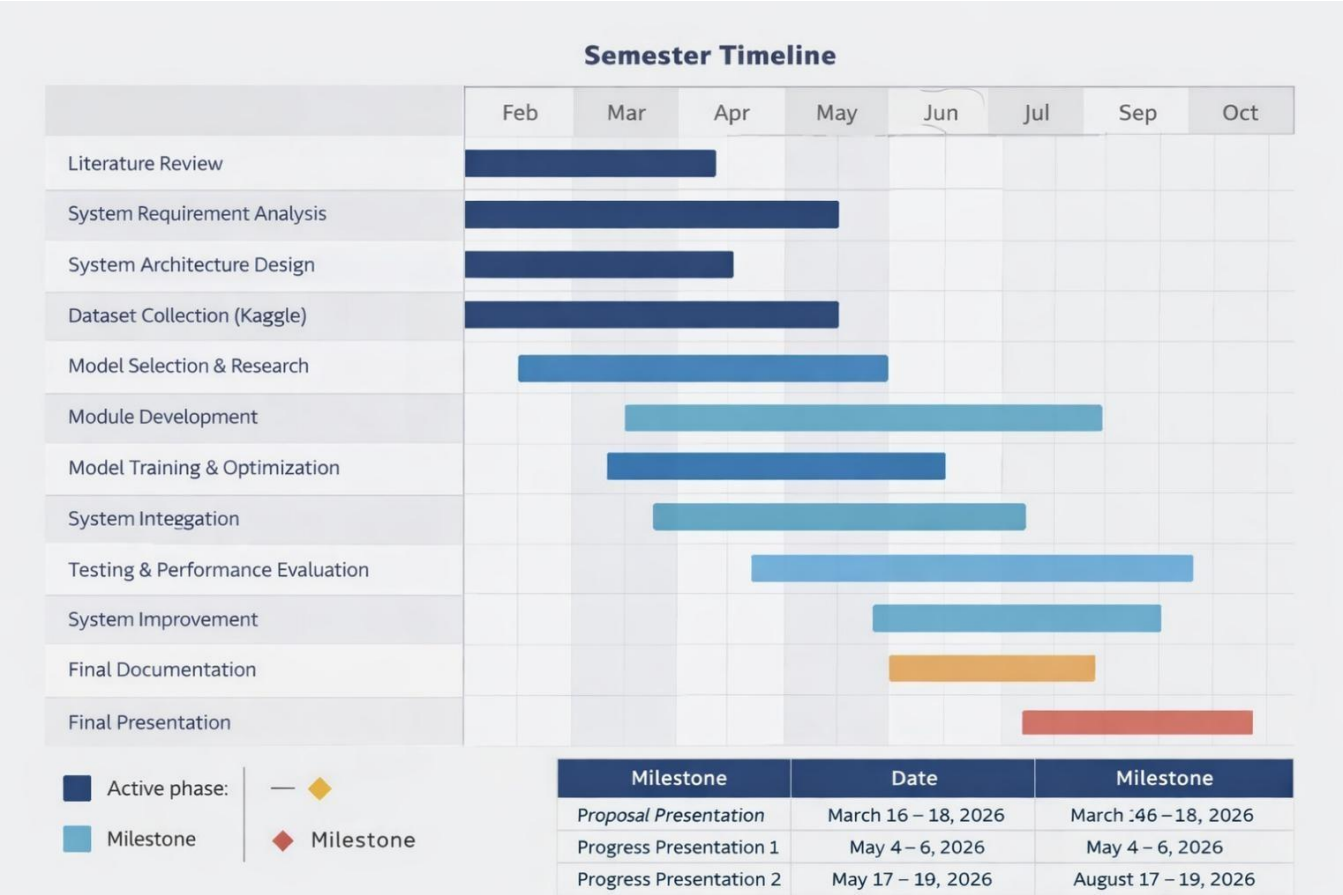


Figure 3: Gantt Chart

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APPENDICES

appendices provide supplementary materials that support the proposed research and system development. These materials include system diagrams, interface designs, risk analysis, and ethical considerations related to the AI-based elder-care monitoring platform.

Appendix A – Survey Instrument

A short survey can be conducted to understand the needs of potential users such as family members and caregivers regarding elder-care monitoring technologies.

Sample Survey Questions

1. Do you currently have elderly family members living alone or receiving caregiver support?
2. How concerned are you about the safety of elderly individuals when caregivers are not directly supervised?
3. What types of incidents worry you the most?
    - Falls
    - Prolonged inactivity
    - Unauthorized caregiver access
    - Medical emergencies
4. Would you feel comfortable using an AI-based monitoring system that analyzes human movement while preserving privacy?
5. Which feature would be most valuable in an elder-care monitoring system?
    - Fall detection
    - Caregiver verification
    - Activity monitoring
    - Alert notifications
6. How important is privacy protection when using camera-based monitoring systems?

The responses from such surveys can help guide system design decisions and feature prioritization.

Appendix B – Detailed System Diagram

The proposed system consists of several interconnected modules that operate together on an edge device.

System Architecture Overview



Figure 4: system architecture Overview

Additional modules in the full system include:

- Face verification module for caregiver authentication
- Human tracking and zone monitoring module
- Schedule monitoring module for daily routine analysis

These components communicate with each other to provide a comprehensive monitoring solution.

Appendix C – Preliminary UI Mockups

The proposed system includes a simple monitoring dashboard for family members or administrators.

Example simplified layout:

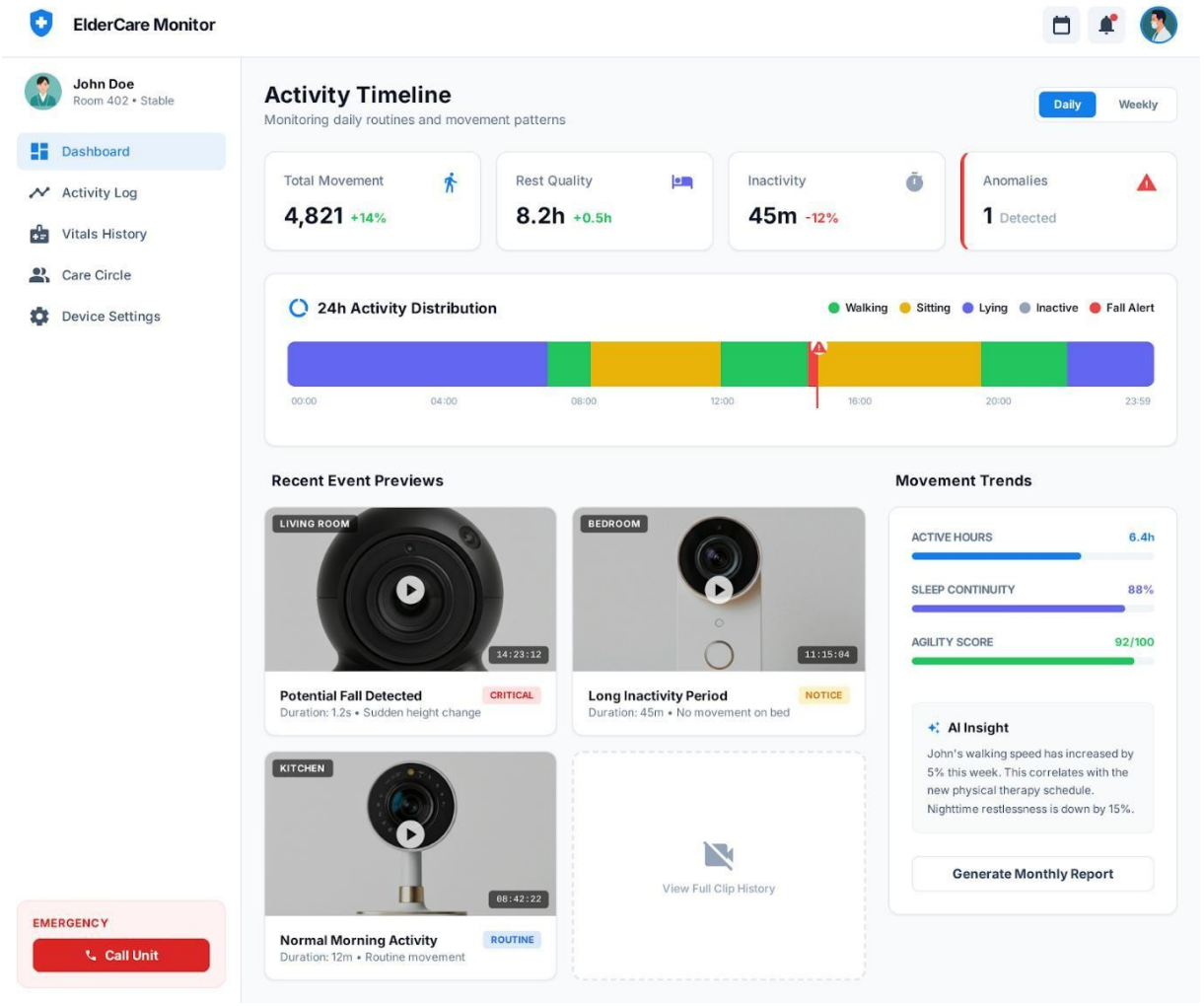


Figure 5: UI mockup1

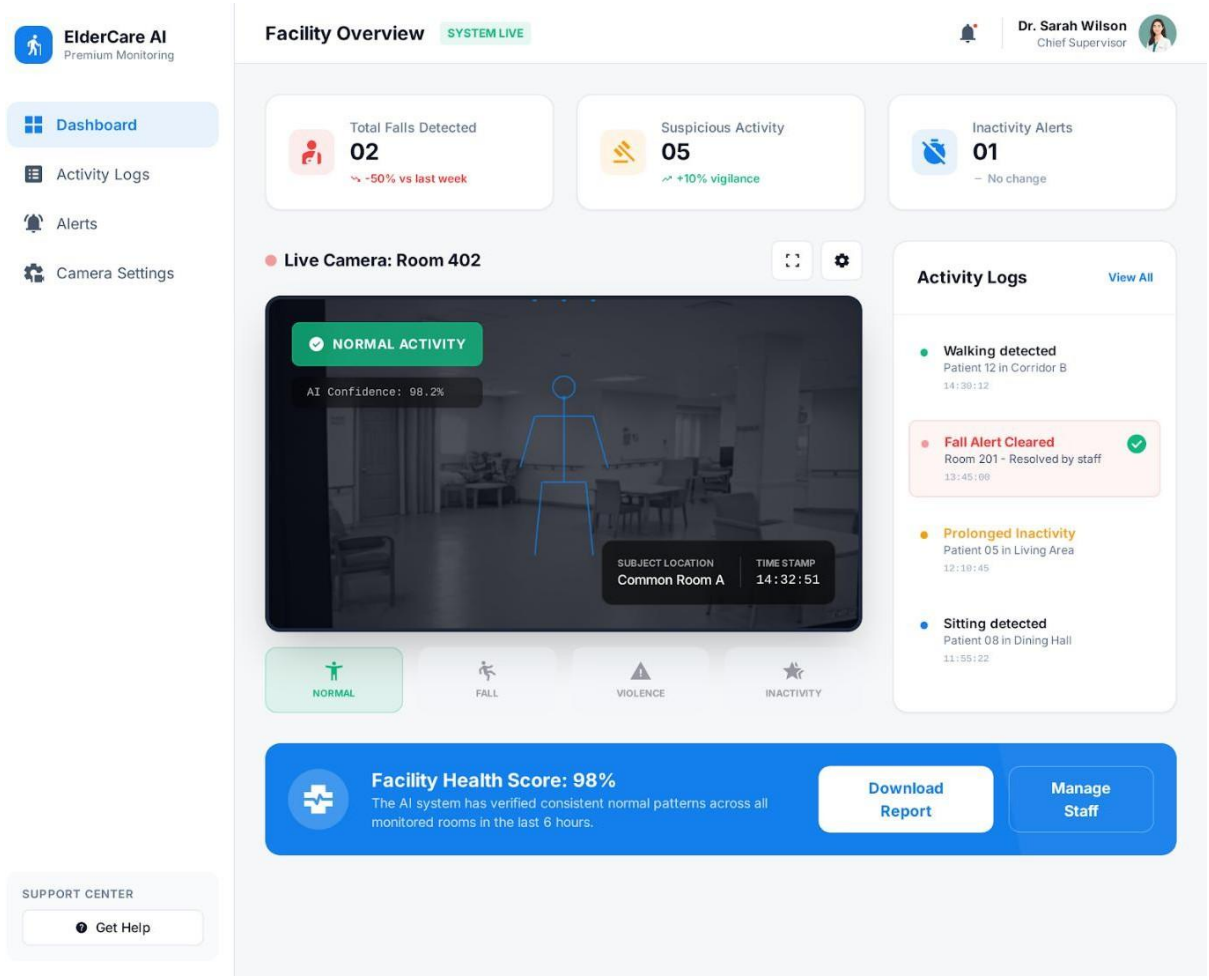


Figure c:UI mockup2

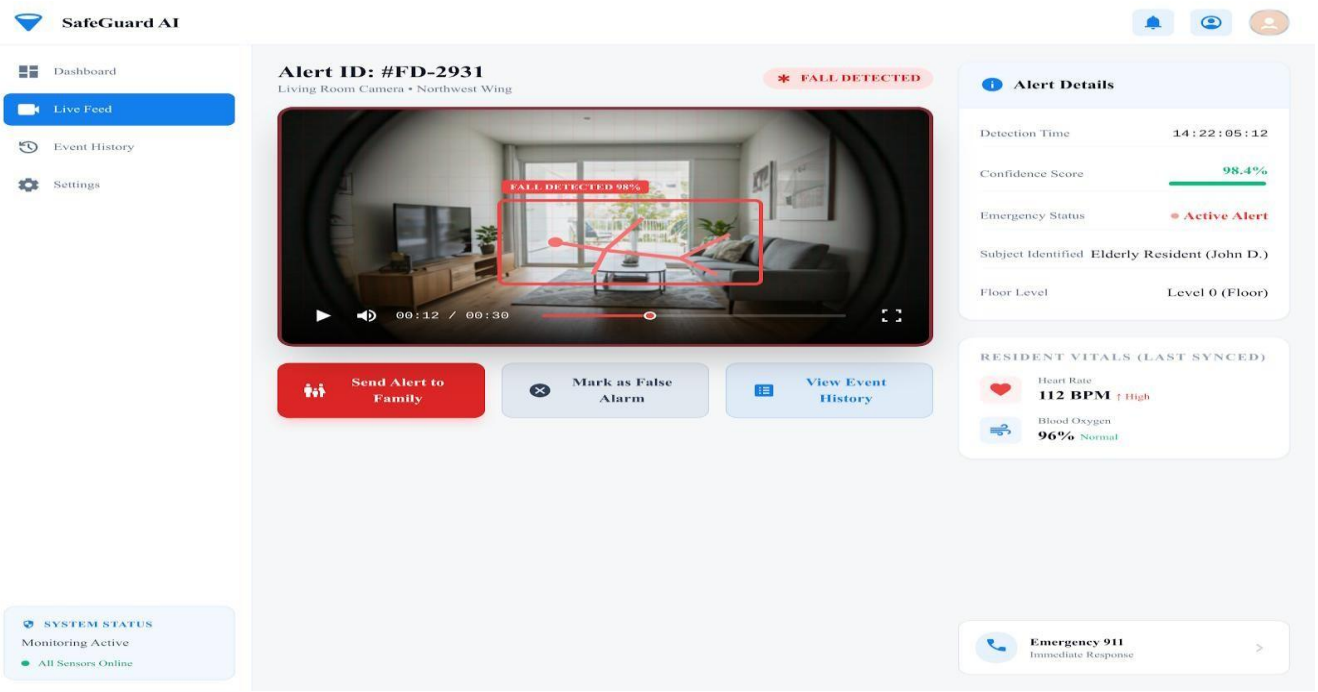


Figure 7: UI mockup3

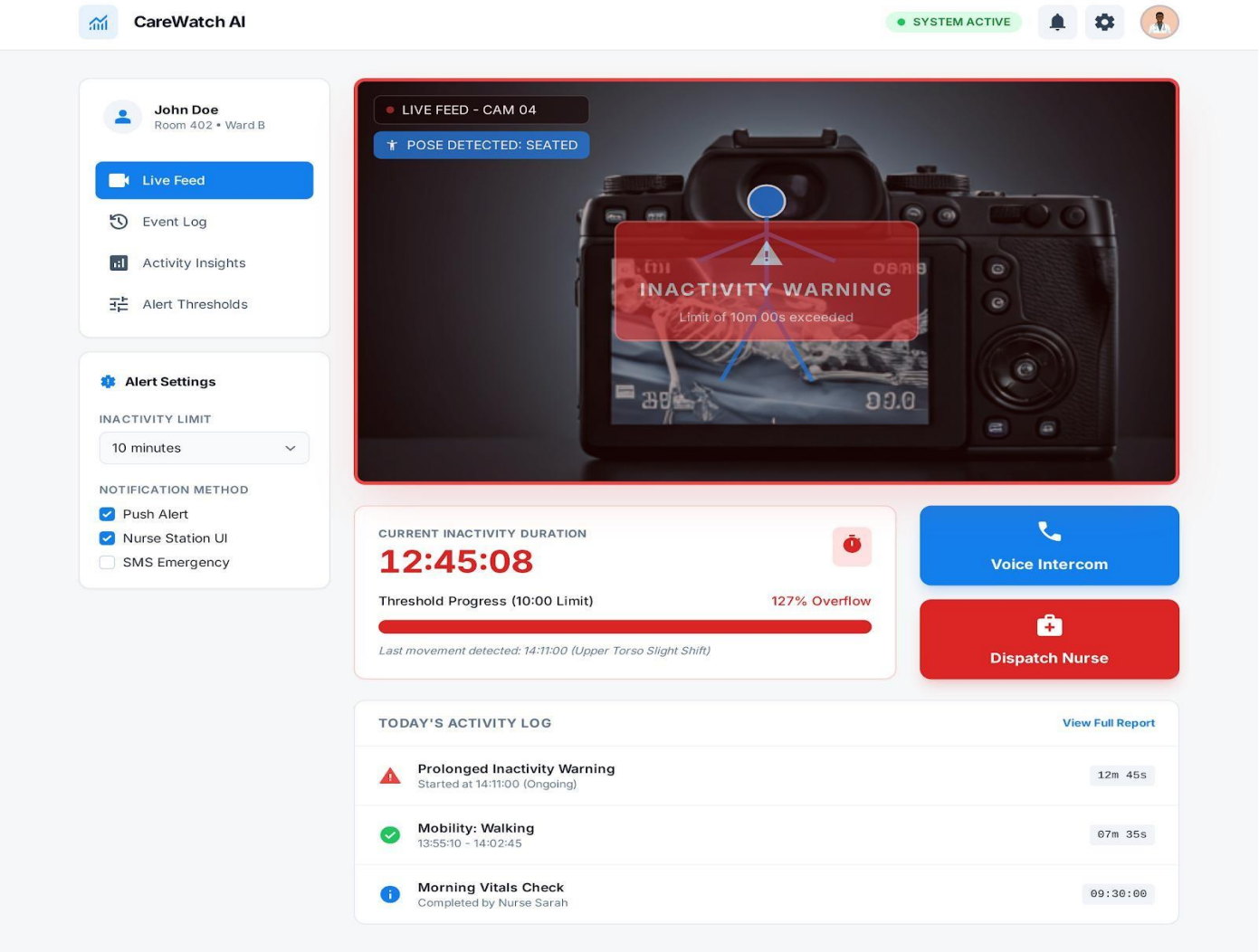


Figure 8: UI mockup 4

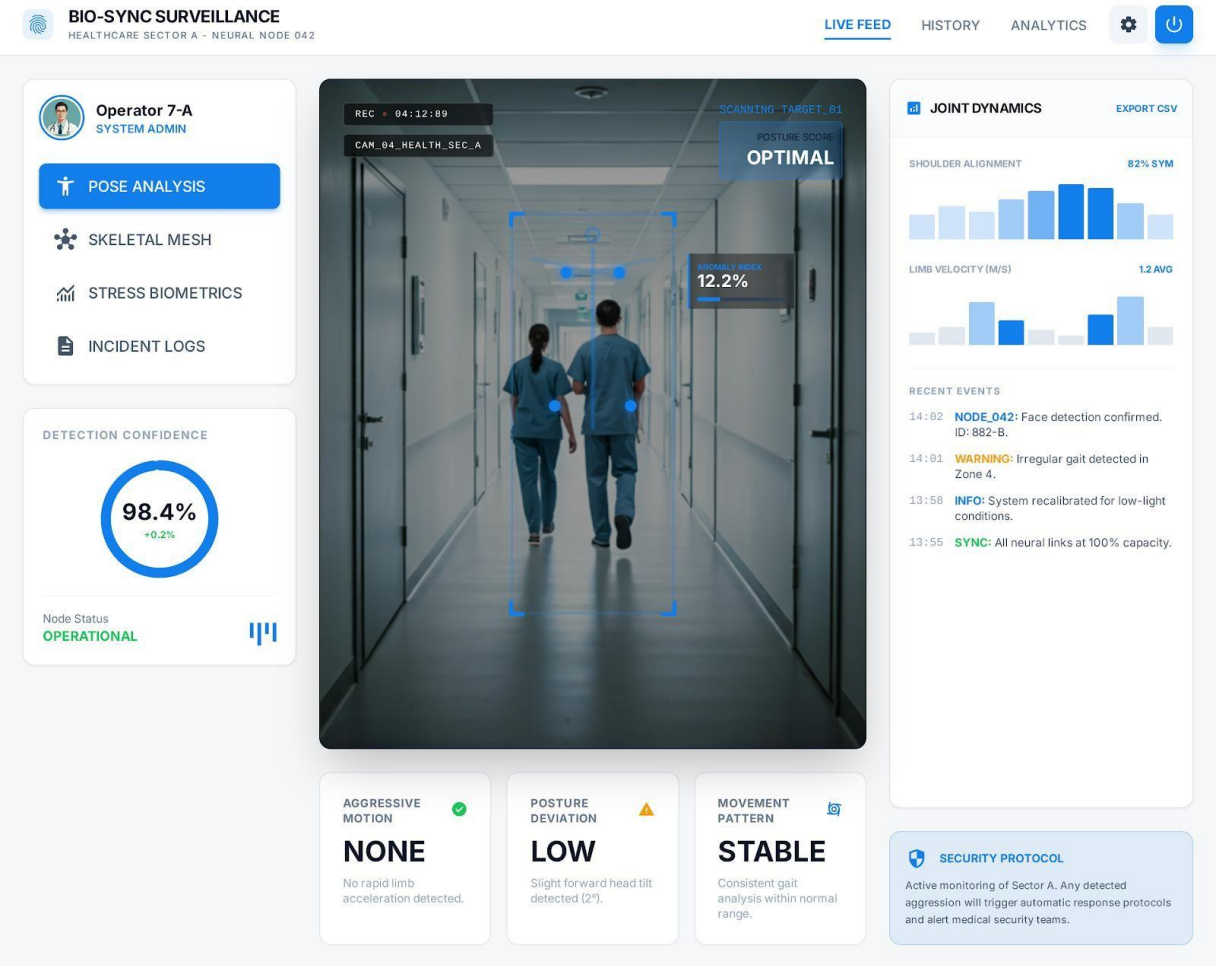


Figure 9: UI mockup 5

Appendix D – Risk Analysis

The following table identifies potential risks associated with the system and mitigation strategies.

Risk	**Description**	**Mitigation Strategy**
False Positive Alerts	The system may incorrectly classify normal activity as an anomaly	Improve model training and adjust detection thresholds
Lighting Conditions	Poor lighting may reduce camera detection accuracy	Use improved camera settings and preprocessing
Hardware Limitations	Edge devices may have limited processing power	Use lightweight AI models optimized for edge computing

Risk	Description	Mitigation Strategy
Privacy Concerns	Users may worry about camera-based monitoring	Use pose-based data instead of storing raw video
Network Issues	Notification system may fail due to connectivity problems	Store event logs locally and retry notifications

Table 5:Risk Analysis

Appendix E – Turnitin Report

